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Here again the MT differs completely from the LXX because two of its words are represented by a rendering reflecting two similar-looking Hebrew words. It seems that initially one of the two words of the MT, בשבי ילכו, was read differently either by the translator or (less likely) by his *Vorlage*.¹ In the new context the "captivity" did not match the "eating", or alternatively the "flesh" did not match the "walking", and therefore the second word had to be adapted to suit the new context (for the "eating", see also the first part of the verse). Thus the translator changed either "captivity" to "flesh" or alternatively "they will walk" to "they will eat".

To summarize, it is in the interest of biblical scholarship to evaluate the merits of each reconstructed variant. Special attention must be paid to a small group of reconstructed variants which bear all the marks of well-supported retroversions, but which probably did not exist in the translators' *Vorlage*.

¹ See p. 175, n. 2.

THE DIACRITICAL DOTS AND THE DEVELOPMENT OF THE ARABIC ALPHABET

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The alphabet from which the Arabic, Hebrew/Aramaic and Syriac scripts were developed was deficient not only in its failure to represent vowels, but also in the fact that its letters were fewer than the consonantal phonemes of any of these languages. These deficiencies were eventually remedied not by the development of new letters, but by the use of small signs – dots, strokes, or miniature letters – placed above, below, or (rarely) between the letters of the text. These signs are generally treated in separate groups: vowel signs, diacritics (i), marking the strong/doubled pronunciation of a consonant or its opposite, and diacritics (ii), distinguishing the different consonants represented by a letter. It is equally reasonable, however, to consider all these signs as marking different features of the sound unit represented by the letter. Diacritics (ii) distinguish the basic consonantal phoneme; diacritics (i) give information on possible variations in its quantity or quality, and vowel signs show it as “at rest” or “in motion” with a particular vowel quality.¹ The signs clearly all derive ultimately from phonological analysis of spoken language, and the little information available on their origin seems to support the view that they should be seen as a single group rather than as three. This paper attempts to show that all these signs are, indeed, the product of the same set of ideas about speech sounds, and that diacritics (ii) – henceforward the “diacritical dots” – provide very interesting information on those ideas.

The letters marked by diacritical dots in the standard Arabic alphabet can be divided, broadly, into two groups: I, those in which the letter is marked by dots above or below to distinguish each consonant it represents; II, those in which the letter is marked by a dot when it represents one of a pair of consonants, but is unmarked when it represents the other. The marking of

¹ The view that the Semitic alphabets were, in fact, syllabaries, is strongly argued by I. J. Gelb in *A Study of Writing* (London, 1952), pp. 147–52. On pp. 152–3 he lists other scholars who have held the same opinion.

jīm, *ḥā'* and *khā'* seems to unite both groups, but for present purposes this problem can be ignored. The pair *jīm* × *khā'* (*|ḥā'*) is provisionally included in group I, and the pair *ḥā'* × *khā'* in group II. The pairs *sīn* × *shīn* and *fā'* × *qāf* as marked in the standard alphabet do not fit these two groups, but in early manuscripts and inscriptions both pairs are marked by the group I method: *sīn* usually with three dots in horizontal line below and *shīn* with the same arrangement of dots above; *fā'* with one dot below and *kāf* with one above (as found also in Western texts).¹ Consequently the groups can be listed as follows:

Group I: *bā'*, *yā'* × *nūn*, *tā'*, *thā'*; *jīm* × *khā'* (*|ḥā'*); *sīn* × *shīn*; and *fā'* × *qāf*.

Group II: *ḥā'* × *khā'*; *rā'* × *ẓāy*; *ṣād* × *ḍād*; *tā'* × *ẓā'*; and *ain* × *ghain*.

The available manuscripts and inscriptions show that the use of (most of) these diacritical dots was already firmly established in the first Islamic century, and remained remarkably stable. The arrangement of the dots in horizontal or vertical line, or triangular group, might vary, but their number, and their position above or below the letter, remained generally the same. The chief exception occurs with *fā'* × *qāf*, for which several different systems of diacritics were used, but other cases of variation are also found. It is assumed here that these variations derive from originally independent local usage, which sometimes became confused with, and was eventually supplanted by, the standard convention.² Such local usage was presumably motivated by particular considerations, phonological or otherwise, but it is not the intention here to attempt to bring these to light. The usage described for groups I and II above is demonstrably early, and in most cases was accepted as standard, so this investigation will be

¹ Examples can be found in N. Abbott, *The Rise of the North Arabic Script* (Chicago, 1939), pp. 38–9, and a fuller discussion in A. Grohmann, *Arabische Papyruskunde* (= *Handbuch der Orientalistik*, erste Abteilung, Ergänzungsband II, erster Halbband), Leiden, 1966, pp. 95–6.

² Variant usage is described in Grohmann, *Arabische Papyruskunde*, pp. 95 f. J. von Karabacek discusses the points on *fā'* and *qāf*, and attempts to determine the period and area of the different systems, in *Sitzungsberichte der Akademie der Wissenschaften in Wien, phil.-hist. Klasse*, CLXXXIV, 3 (1917), 25–8. He is probably not right, but Grohmann's remark that all systems noted by Karabacek appear in Egypt (*From the World of Arabic Papyri*, Cairo, 1952, p. 85) does not rule out the possibility of different local origins for the different systems. All three systems of Hebrew pointing were, after all, also used in Egypt.

concentrated on it, leaving local variations for consideration elsewhere.

The standard Arabic alphabet also makes use of other signs which can be considered as diacritics on the same level as those discussed above. These can all be classed in a group III, which would include *hamza*, the dots on *tā' marbūṭa*, and the mark in final *kāf*. The motivation for these signs is obvious: the sign itself indicates the sound required. Thus the mark which distinguishes final *kāf* from final *lām* is a miniature *kāf*, as is clear in early manuscripts. The *hamza* sign, used on the letters *hā'*, *wāw* and *yā'* when they represent this consonant, is derived from 'ain, to which its sound was likened.¹ The two dots which mark *hā'* as *tā' marbūṭa* are those of *tā'* itself. Other such signs are widely used in literary texts, where the use of unusual words and poetic language increased the problems of the reader and made the accurate distinction of the consonants more than ever a necessity. For instance, a letter representing *ḥā'*, 'ain or *ṣād* (which is unmarked in the standard system) is shown to represent the sound in question by a miniature form of the same letter, placed above or below it. Thus the letter itself is used to characterize the sound which it most commonly represents.² These signs are, however, a supplement, not an alternative, to the standard system, and represent the latest layer of diacritics to be applied to the Arabic alphabet.³ The introduction of these letter-form signs as conso-

¹ E.g. by 'Uthmān al-Dānī, "The place of *hamza* in a word can be tested by 'ain. Wherever 'ain comes (in pronunciation), there is the place of *hamza*." *Kitāb al-Naqt*, ed. O. Pretzl in *Orthographie und Punktierung des Koran* (Istanbul, 1932), p. 150, l. 13.

² There is an unusually complete example of such usage in N. Abbott, *Studies in Arabic Literary Papyri*, III (Chicago, 1972), text 6 (p. 149). Here, as elsewhere, the unmarked member of some group II pairs is given a sublinear dot, as described by A. Grohmann, in *Allgemeine Einführung in die arabischen Papyri* (Wien, 1924), p. 72. Such signs were necessary because scribes did not mark the standard diacritical dots consistently. A special sign (*muhmal*) indicating that a letter was correctly left undotted eventually came into standard use. The function of these "differentiating signs" (as Grohmann calls them) is the same as that of the standard dots: to indicate the particular consonant represented by the letter. The confusion in the marking of *sīn/shīn* noted there is perhaps to be explained by the fact that the signs used (a stroke, or one or three dots, usually above the letter) are (in some cases at least) intended to identify the letter (which in an undotted text could be easily confused with others) but not the particular consonant represented.

³ Thus B. Moritz remarks that *tā' marbūṭa* appears to have been the last letter to receive its dots – in the second century of the Hegira. *Encyclopaedia of Islam*, 1st edition, art. "Arabia. d. Arabic Writing".

nantal diacritics, and also as signs for vowels and *tashdīd* as in the standard system is attributed to al-Khalil.¹ We have here, then, a clear case of a uniform approach to all three types of vowel and diacritical signs which seems to support the suggestion, made above, that they were not considered as marking phenomena fundamentally different in nature. This is perhaps not surprising in al-Khalil's situation, however, and our interest is mainly with the earlier period.

The consonantal diacritics of group II represent nothing more than the easiest way of distinguishing two sounds represented by the same letter. The letter is marked when it represents one of the sounds, but not when it represents the other. No information about the sounds themselves is given. Economy of effort obviously requires that (in contrast to the group III method where the sound marked by a diacritic must be that most characteristically represented by the letter) in group II the less common of the two sounds is represented by the marked form of the letter. This is a simple and effective method, but in the difficult literary texts it was found necessary to supplement it with signs from group III, particularly since scribes often failed to mark the standard diacritical dots.

Once the group has been identified, the motivation for the use of consonantal diacritics in group I also seems quite clear. It is based on the physical production of the sounds represented, following the view that progression from the back of the mouth to the front was progression downwards. A sound which is articulated towards the back of the mouth is "high" (and so marked with a supralinear dot) in contrast with a sound articulated towards the front of the mouth, which is "low" and marked with a sublinear dot. This view is best known from the description of the vowel system of Hebrew by Saadya Ga'on and from its use in the Masora. It is also, however, the basis of the East Syrian system of vowel signs, and of the system taken over by the Arabs, as is shown by the vowel names *naṣb* "raising" and *khafḍ* "lowering".² In group I, as in group III, then, the diacritics are based on the consonantal sounds they represent.

The application of this view to the consonants of Arabic was

¹ See N. Abbott, *Studies in Arabic Literary Papyri*, III, 7.

² Arab grammarians offered other explanations. See E. W. Lane, *An Arabic-English Lexicon*, Book I, part 8 (London, 1893), p. 2799, col. 2, but it seems more likely that the names were taken from Syriac *ṣeqāfā* "raised" and *rebāṣā* "depressed" (or similar terms) along with the signs.

probably based on an arrangement of the alphabet such as that of al-Khalil. He lists the Arabic consonants in linear order according to their position of articulation (*makbraj*), starting at the back of the mouth and working forwards.¹ An arrangement of this sort would allow the distinction of any two consonants on the basis of their *makbraj*. The type of presentation more typical of Arabic grammarians (and also used by al-Khalil), which lists "articulation groups" of several consonants produced (in different ways) from the same *makbraj* could only serve as the basis for the distinction of members of different articulation groups. However, an arrangement of this sort could have been the basis of the rather small number of distinctions marked by the group I diacritical dots (see note 2 below). Al-Khalil also shows the basis for the use of sublinear dots to mark *yā'*. For him, *yā'* is not a palatal consonant, but is one of the *hawā'īya*, which are considered not to have any *makbraj* within the mouth, and so to be really "extra-buccal". Hence *yā'* is placed at the end of al-Khalil's list, as even further forward than the labials. The diacritical dots were clearly based on the same view. The dental/alveolar *nūn*, *tā'* and *thā'* are marked with supralinear dots as "high" or "back" sounds in contrast to the "bilabial and beyond" *bā'* and *yā'*. Similarly pharyngeal *khā'* (*/ḥā'*) contracts with palatal *jīm*, palatal *shīm* with dental *sīm*, and palatal *qāf* with bilabial *fā'*. In this respect, then, the view of the Arabic consonants presented by al-Khalil was the same as that which had been used, some time before, as the basis for the group I use of diacritical dots.

One question must still be considered. Why are some pairs of consonants distinguished by the group I method when the simpler method of group II would have sufficed? The problem is, indeed, emphasized by the fact that *sīm* and *shīm*, which were originally distinguished by the group I method, are, in the standard alphabet, members of group II (with a supralinear sign composed of three dots, instead of one). It seems probable that the group I method was used only to distinguish consonants widely different in *makbraj*.² Thus the *makbraj* of palatal *jīm* is

¹ This list is given in the introduction to the *Kitāb al-ʿAin* (ed. A. Darwish, Cairo, 1968, pp. 53, 65). A form of this introduction is translated by J. A. Haywood in *Arabic Lexicography* (Leiden, 1960), pp. 28-37. Al-Khalil's views on phonology are discussed by S. Wild in *Das Kitāb al-ʿAin und die arabische Lexicographie* (Wiesbaden, 1965), pp. 29-35.

² It is, then, possible that only sounds occurring in different articulation groups were distinguished, but no known list of articulation groups fits this

distant, and quite distinct, from that of pharyngeal *ḥā'* and *khā'*.¹ Similarly the dental/alveolar *makbraj* of *nūn*, *tā'* and *thā'* is distinct from the "bilabial and beyond" position of *bā'* and *yā'*,² and so also are the positions of *qāf* and *fā'*, and of *sīn* and *shīn*. In contrast, the pairs of sounds distinguished by the group II method are articulated in similar or identical positions. Thus the points at which *ḥā'* and *khā'* are articulated are very close, and the same is true for *ʿain* and *ghain*. The positions of *ṭā'* and *ẓā'*, and of *ṣād* and *ḍād* are less close – even quite far apart according to some – but they are pronounced in the centre of the mouth, where obviously distinct articulators are lacking, so the failure to treat the members of these pairs as distinguished by a major difference in *makbraj* would not, in any case, be surprising. A similar explanation can cover *rā'* and *ẓāy*. These sounds are, indeed, far apart in al-Khalil's list. For Sibawaih, however, the articulators of *rā'* and *ẓāy* are very close,³ and in the standard Hebrew listing the two sounds occur in the same articulation group (see below). If this view is correct, then, it would seem that group I, based on major differences in *makbraj*, represents the first attempt to develop an alphabet suitable for Arabic from the sixteen or so letters available. Group II, making finer distinctions, represents a second stage in this work, and group III represents the finishing touches.

As has been pointed out, the ideas on which al-Khalil based his arrangement of the alphabet are of Indian origin, and are exemplified in the order of the Sanskrit alphabet.⁴ This must also

situation. The closest is the Hebrew list given below, but there *sīn* and *shīn* are placed in the same group.

¹ This fits the assumption, made above, that initially *jīm* was distinguished from *ḥā'*/*khā'*, and that the distinction of *ḥā'* from *khā'* was introduced later.

² Al-Khalil assigns *nūn*, *tā'* and *thā'* to three different articulation groups, but in the Hebrew and Syriac lists given below they belong to the same group. It is assumed that originally the letter was marked with a supralinear dot when it represented *nūn*, *tā'* or *thā'*, and with a sublinear dot when it represented *bā'* or *yā'*, and that further differentiation came later.

³ *Zāy* is articulated with the tip of the tongue, *rā'* with the part a little behind the tip. The other articulator "above the incisors" is the same for both. See Wild, *Das Kitāb al-ʿAin*, pp. 92–3. The descriptions attributed to al-Khalil are given for comparison, and suggest that for him also the *makbraj* of *rā'* was in fact close to that of *ẓāy*.

⁴ See Wild, *Das Kitāb al-ʿAin*, pp. 37–40. He concentrates on al-Khalil's articulation groups. J. A. Haywood, *Arabic Lexicography*, p. 37, points out the dependence of al-Khalil's arrangement of the alphabet on Indian sources, and that, in the present context, is probably equally significant.

be true of the ideas on which the diacritical dots are based. Arabic tradition states that the diacritical points were borrowed from the Syrians. This cannot be true of the signs themselves, since the requirements of Arabic were quite different from those of Syriac. If the tradition is based on anything more than the mere fact that the Syrians had an older tradition of written literature than the Arabs, it must refer to the theory on which the use of the points was based. There are, in fact, some features of the Syrian pointing – and also that of Hebrew – which might derive from this same Indian influence. It is possible, then, that the Arabic tradition reflects the fact that Indian ideas were transmitted to the Arabs by Syrians or Jews.

The only two consonants which are distinguished by diacritics in the Syriac alphabet as normally presented are *resh*, marked with a dot above the letter, and *dālath*, marked with a dot below the same letter. In addition, however, the six letters *b g d k p t* can represent either stops or fricatives. The stop is represented by a dot above the letter, the fricative by a dot below. The signs with the corresponding function in some Hebrew MSS with Palestinian pointing can be interpreted as pointing upwards (*dagesh*, marking the stopped pronunciation), or downwards (*rafe*, marking the fricative), although both signs are placed above the letter. The corresponding signs in the Tiberian pointing do not visually suggest the idea of “above” or “below”, but in some contexts in the Masora *dagesh* is a synonym of *millāʿel* “above” and *rafe* of *millāraʿ* “below”.¹ The terms *millāʿel* and *millāraʿ* are most familiar to Hebrew scholars as designations of accent or stress position, where *millāraʿ* “below” designates stress on the final or “left-hand” syllable, while *millāʿel* “above” designates stress on the penultimate – or, in a two-syllable word, the initial or “right-hand” syllable. One case is known in a Tiberian Masoretic text in which, in connection with these terms, dots are used above the syllables in question to mark the position of the stress.² Dots are similarly used to distinguish *sin* from *shin*. A dot above the left-hand side of the letter (corresponding to the left-hand accent position, *millāraʿ* “below”) marks *sin*, and a dot above the right-hand side (and so *millāʿel* “above”) marks *shin*. As far as I know, the terms *millāraʿ* and *millāʿel* are not used in connection with the dots on *sin* and *shin*, but this interpretation of the graphic

¹ See S. Morag, “Some Aspects of the Methodology and Terminology of the Early Massorettes” in *Lesbonenu* xxxviii (1973-4), 68-9.

² See *ibid.*, 63, note 35.

symbolism is supported by the fact that, in at least one MS with Palestinian pointing, the sign marking *sin* is placed below the letter, that marking *shin* above.¹

In addition to these features of the pointing systems, which, at least in the case of the Syriac items, were in use in the early seventh century, there is a certain amount of relevant information in grammatical writings. The consonants of Hebrew are traditionally classified in five groups, according to position of articulation, and described, starting with those produced farthest back in the mouth, as letters of the throat ' *h ḥ* ', palate *g y k q*, tongue *d ṭ l n t*, teeth *ṣ ṣ ṣ r ṣ* and lips *b w m p*. This list goes back to the eighth century or before.² Similarly the Syrian grammarian Elias of Soba (c. A.D. 1000) lists the consonants – also in five groups – “in order according to the natural pronunciation of the consonants” ‘ *h ḥ* ’ *g k q* : *w y l* : *ṣ ṣ ṣ r* : *d ṭ t n* : *b m p*.³ It is interesting to note that this grouping is not fully consistent with the more detailed description of the articulation of the consonants which precedes it, and which is undoubtedly based on Arabic models. This listing shows that the Syrian grammarians felt it

¹ See Revell, *Hebrew Texts with Palestinian Vocalization* (Toronto, 1970), p. 87 (TS 16: 96). The sign marking *sin* is placed below the letter in various other MSS, but (at least in the small fragments available for study) the method used in these must be classed with group II (*shin* has no diacritic) or III (*samek* and *shin* are used as diacritics), and in these groups the position of the sign is not significant.

² The earliest known source for this list is the *Sefer Yeṣira*, which is generally dated to the eighth century, though placed earlier by some. Allony also dates this list to the eighth century, on the assumption that the phonological knowledge on which it is based, which must have originated in India, could only have come to the Jews through the Arabs (see *Sinai* LXXIV, 1974, 49 f.). However, it is possible that the knowledge could have come through other channels, and it seems unlikely that the *Sefer Yeṣira* contains the first formulation of the list. The work is concerned with cosmology and cosmogony, not language, and the groups are arranged not according to their positions of articulation, but to suit the author's purpose, with their initial letters in alphabetical order. (See the opinion of Saadya Gaon in his commentary on *Sefer Yeṣira* iv. 3 in M. Lambert, ed., *Commentaire sur le Sefer Yeṣira... par le Gaon Saadya* (= *Bibliothèque de l'école pratique des hautes études, Sciences philologiques et historiques*, fasc. 85, Paris, 1891), text p. 75. Consequently it would seem that the list must have been produced by the eighth century, and could have been produced earlier.

³ See *A Treatise on Syriac Grammar by Mari Elia of Soba*, ed. R. J. H. Gottheil (London, 1887), text p. 7. Despite his assertion, the order of the letters within the groups clearly owes something to the alphabetical order.

not only important to describe the physical production of the sounds of their language, but also to arrange them in linear order according to the point of articulation, just as was done by al-Khalil. The order in which the sounds are given shows a number of differences from that of al-Khalil, however, so it seems quite possible that Elias is here presenting an independent Syrian tradition, not one derived from the Arabs. Both Syrian and Hebrew lists are divided into five groups, and in the case of the Hebrew list not only the number, but even the names of these articulation groups recall the five major groups of Indian consonants: velar, palatal, "lingual", dental, labial.¹ Allony has pointed out that this list is based on ideas originating in India, but considers that the Jews acquired them from the Arabs. This is, of course, possible, but by al-Khalil's time there was already a long history of communication between the Near East and India so it is also possible that the Hebrew list, and also the Syriac, was based on Indian thought which did not come by way of Arabic scholars.

According to Elias of Soba, the Syrians pronounced *resh* further back in the mouth than *dalet*. This would justify its being marked as "higher" by a supralinear diacritical dot, in contrast to the "lower" *dalet* which had a sublinear dot. This appears to be a coincidence, however, as the earliest Syriac writing uses the dot only with *resh*, and not with *dalet*, following the group II

¹ These terms – save for "lingual" – are those used by W. S. Allen in *Phonetics in Ancient India* (London, 1953). The "velar" series is said by the Indians to be pronounced at the "root of the tongue" (Allen, p. 51) and the Hebrew "throat letters" "at the far end of the tongue" in the *Sefer Yešira* (*sof*) and Saadya's (somewhat expanded) translation (*'aqsā*) (ed. Lambert, p. 74). The usual Indian term for the third series is obscure, giving rise to terms like "cerebral" and "domal" in European grammars. Allen calls it "retroflex" because it is characterized by retroflexion of the tongue rather than by a place of articulation (Allen, p. 52). The same consideration (presumably) has given rise to the term "lingual" in some European grammars. Wild suggests that the term *inḥirāf* applied by al-Khalil to *rā*, *lām* and *nūn* is simply a translation of this Indian terminology (*Das Kitāb al-'Ain*, p. 32, n. 27 and p. 40). It would not be surprising if other Near Eastern scholars, also copying the Indian tradition, should also base the name for their third group of consonants on this tongue movement, for the tongue is, after all, an articulator, and the traditional Indian name does not seem to indicate a place of articulation. Wild sees al-Khalil's articulation groups as derived from the same Indian tradition, and cites his listing of eight groups as indicative of this. However, these eight groups exclude the four *ḥawā'īya*, while the eight Indian groups include the semivowels. In some respects, then, the Hebrew listing seems closer to the Indian than does al-Khalil's.

method already found in the Nabataean alphabet.¹ The available information certainly does not suggest that the six stops *b g d k p t* were pronounced farther back, or "higher" than the corresponding fricatives, so the position of the dots distinguishing stops from fricatives cannot depend on position of articulation.² The traditional listings do not, however, include both stops and fricatives, and the interest in their different positions of articulation was certainly a product of Arabic influence. However, in the Sanskrit alphabet stops are listed before other varieties of consonant (aspirates, nasals), and so would be "higher", and the Syrian and Hebrew traditions may have imitated this. The marking of *sin* and *shin* in Hebrew could possibly have been copied from Arabic, but if this were so one would expect the Jewish scholars to copy the Arabic signs as well as the idea. It seems more likely that both Jews and Arabs derived the idea from a common source – the knowledge of the Sanskrit alphabet and its phonological basis which had been brought to the Near East from India.

It is possible that the view common to Jewish and Syrian grammarians that the back vowels were "high" in contrast to "low" front vowels also reflects Indian influence. This idea seems to be symbolized graphically by the use, in Syriac, of a supra-linear dot to distinguish a word characterized by a vowel produced towards the back of the mouth from a homograph characterized by a vowel produced more towards the front, which was marked by a sublinear dot.³ However, according to Jacob of Edessa, who gives the earliest description of the basis for this classification of vowels, the back vowels are "hard" or "thick" (*'be'* or *'pte'*), while the front vowels are "thin" or "clear" (*qaṭīn* or *nqed*), and the same terms are applied to consonants, where

¹ The dot below *dālath* was still optional in the fifth century. See J. B. Segal, *The Diacritical Point and the Accents in Syriac* (London, 1953), p. 11, n. 3.

² For Bar Hebraeus the two pronunciations of *g d k t* were evidently (much) the same as Arabic *j gh, d dh, k kh, t* and *th*, and the stopped pronunciation of *b* and *p* was bilabial (as Arabic *b*), while the fricative pronunciation was labiodental (as Arabic *f*). See his description in *Le livre des splendeurs* (ed. A. Moberg, Lund, 1922), pp. 193–4 (translated, *ibid.*, *Buch der Strahlen*, part 2, Leipzig, 1907, pp. 7–8). The situation, at least for *d, k* and *t*, was evidently the same for Saadya. See M. J. Dérenbourg in *Journal Asiatique*, 6ème série, xvi (1870), 518.

³ But note that it was not necessary to mark o/u vowels in this way, so that *zeqāfā* (ز) was the farthest back of the vowels to be so marked. See J. B. Segal, *The Diacritical Point*, pp. 20 f.

they cannot refer to position of articulation.¹ Consequently there is no reason to suspect Indian influence. The listing of the vowels in the order in which they are produced, "descending" from the back to the front of the mouth appears only in treatises by Saadya Gaon and later writers, and in certain Masoretic lists.² These lists distinguish homographs in the same way as described for the dots in Syriac. A word characterized by a back vowel is described as "high" (*millā'el*) in contrast to a homograph characterized by a vowel produced more towards the front, which is "low" (*millara'*). The distinctions made in this way are quite narrow, e.g. *holem* is distinguished from *gameš*, and *gameš* from *pataḥ*, and it does not seem possible that they could have been based on any phenomenon other than the position of origin of the vowels in the mouth, envisaged in much the same way as described by Saadya. This view of the vowel system was not typical of the early Jewish and Syrian grammarians,³ and is therefore likely to have derived from foreign influence – ultimately Indian.

¹ For the terms applied to vowels, see G. Phillips, ed., *A Letter by Mar Jacob Bishop of Edessa on Syriac Orthography* (London, 1869), text p. 14 and for consonants see A. Merx, *Historia artis grammaticae apud Syros* (Leipzig, 1889), text p. 78. *g* and *d* are "thick", *k* and *t* "medium", and *q* and *ṭ* "clear". The terms presumably refer to some quality such as "sonorousness" which was recognized as present in *g*, *d* and back vowels to a much greater extent than in *q*, *ṭ* and front vowels. However, it is significant for this paper that both vowels and consonants were classified on the same basis.

² See the edition of Saadya's treatise on the vowels by S. Skoss in *J.Q.R.* XLII (1951-2), 283 ff., and for the "descending" (*ḥaṭṭa* or *ḥabaṭa*) of the vowels in order from back to front, pp. 302 ff. A Masoretic list of the sort described is given in S. Frensdorff, *Okblab we-Okblab* (Hanover, 1864), list no. 5. The order of vowels on which these lists from the "Tiberian" Masora is (from "high" to "low") *u o ɔ a e e i*. Lists of vowels ascribed to "the Tiberians" begin with *ɔ* or *a* and end with *o* or *u*, although the order of individual vowels varies. See Allony, *Beth Mikra* LIII, 2 (1973), 251 (lists ascribed to the Tiberians) and *Lesbonenu* XXIX (1968), 140-2 (treatise ascribed to Moshe ben Asher). Since the question of Arabic influence on Hebrew has been raised, note that the vowel listing showing Arabic influence is the tripartite one: *rum* (= *raf*) *o*, *u*, *maṭṭa* (= *ḥaḥaf*) *e*, *i* and *leḥaṣṣiv* (= *naṣṣiv*) *ɔ*, *a*, *e*. See e.g. *Die Diquḡe ha-Te'amim*, ed. S. Baer and H. L. Strack (Leipzig, 1879), no. 36, and cf. *Journal Asiatique*, 6ème série, XVI (1870), 364.

³ For Jacob of Edessa's order see A. Merx, *Historia artis grammaticae*, text pp. 80 ff., where noun bases are listed with characteristic vowels in the order *a*, *e*, *i*, *ē*, *o*, *u*, *ū* (with Bar Hebraeus' names, *petāḥā*, *rebāṣā*, *ḥebāṣā arīkā*, *rebāṣā karyā*, *'eṣāṣā arīkā*, *'eṣāṣā karyā*). The MS is broken at both ends of this list, but it is highly probable (considering other lists of Syriac vowels) that *ɔ* (*zeqāfā*) stood at the beginning. Thus this is probably based on the Greek

The method of distinguishing homographs used in the Masoretic lists described must have originated before the introduction of vowel signs, and so must go back at least to the sixth century. It seems likely, then, that some knowledge of Indian phonological science – at least the knowledge of the traditional order of the Sanskrit alphabet and its phonological basis – reached the Near East in the sixth century or before. This was combined with the rather limited views on phonology already held there, and gave rise to the idea that sounds produced towards the back of the mouth were “high” in contrast to those produced towards the front. In Hebrew and in Syriac, this concept was applied mainly to the distinction of vowel sounds, because in these languages the consonantal phonemes were adequately represented in the script, so that homographs generally had to be distinguished on the basis of their characteristic vowels.¹ In Arabic, however, the number of consonants was nearly twice as great as the number of different letters in the script (designed for Aramaic or Syriac) which was being used to write it, so that the immediate requirement for the distinction of homographs was some method of identifying further consonants. The theory of “high” and “low” sounds was applied to this purpose, and the group I method evolved. In this first stage, apparently, only broad distinctions in position of articulation were indicated, so that a number of cases remained in which two or even three consonants were represented by the same letter. The use of the group II method to distinguish these evidently represents a second stage in the development of the Arabic alphabet. The differentiation of *nūn*, *tāʾ* and *thāʾ*, and of *bāʾ* and *yāʾ*, by different numbers of dots was probably also introduced at this stage. The extra dots seem to have been added according to the order of the *abjad* (or the Hebrew/Syrian alphabet). The whole process was already complete in the first Islamic century, when most if not all of the diacritical dots are found in use.

In contrast to the Syrian/Jewish theory which gave rise to the group I method of distinguishing consonants, al-Khalil describes sounds produced towards the front of the mouth as “higher”

order with *ʾ* and *a* corresponding to *alpha*, *i* to *ēta*, *ē* to *iota*, and *u* and *ū* to *upsilon*. Other lists give the Syriac vowels in various orders, but typically begin *ʾ a e* or *ʾ e a*.

¹ The basic concern of early efforts to ensure correct reading of written texts was to distinguish homographs. The attempt to indicate pronunciation accurately represents a later stage. Cf. Morag, *Leshonenu* xxxviii, 63 f.

than those produced towards the back.¹ It is probable, then, that his ideas derive from a different source, perhaps closer to the Indian origin of the material. The group I method of distinguishing consonants must have been the work of Christians or Jews who began to use Arabic in literary texts, very likely in Iraq or Syria.² Once the theory had served its purpose, it was likely forgotten, and never passed on to adherents of Islam. The diacritical dots based on this theory remain in use, however, a testimony to the accurate analysis of speech sounds by the scholars who began the development of the letters used for Aramaic or Syriac into the Arabic alphabet.

¹ *Arfa'* (*Kitāb al-'Ain*, ed. Darwish, p. 64). This might derive from the view (held by the Indians, Allen, *Phonetics*, p. 21, but obviously likely to be arrived at by anyone considering the problem) that the airstream which produces speech sounds rises through the body. It is, however, noteworthy that Saadya also held this view (see Skoss, *J.Q.R.* XLII (1951-2), 292-3).

² For the probable literary use of Arabic in pre-Islamic times, see N. Abbott, *Studies in Arabic Literary Papyri*, II (Chicago, 1967), 5.